

Systematic Verification of the Intuitionistic Modal Logic Cube in Isabelle/HOL

Hanna Lachnitt, Maximilian Claus, and Christoph Benzmüller

Dep. of Mathematics and Computer Science, Freie Universität Berlin, Germany
h.lachnitt@fu-berlin.de, m.claus@fu-berlin.de, c.benzmueller@fu-berlin.de

Abstract. Due to its computational interpretation there has been a lot of interest in intuitionistic logic in computer science. Adding combinations of the intuitionistic modal axioms to intuitionistic modal logic IK results in different systems. Together they constitute the intuitionistic modal logic cube. We use an embedding of intuitionistic modal logic in higher order logic to verify this cube. Automatic reasoning tools, such as Sledgehammer and Nitpick, were used to prove the inclusion relations between the cube's logics and the equality of some logics.

Keywords: Intuitionistic Modal Logics · Frame Correspondences in Intuitionistic Modal Logics · Isabelle/HOL

1 Introduction

We present a formal verification of the intuitionistic modal logic cube, which is a hierarchically structured collection of axiomatisations of different intuitionistic modal logics. This paper can be seen as a straightforward extension of [4], in which the structure of the classical modal logic cube is verified, to the intuitionistic case.

Our investigation is motivated by the proclaimed usefulness of intuitionistic modal logics (IML) in computer science. For instance, Alex Simpson lists several application fields in his dissertation [2], including staged computation, computational effects, security, distributed computation, and typed lambda calculi.

Essentially, IML refers to several extensions to standard intuitionistic propositional logic, featuring the common $\Diamond$ and $\Box$ modalities. The meaning of these modalities is fixed by the following axioms, called k-axioms:

$$\Box(A \rightarrow B) \rightarrow (\Box A \rightarrow \Box B) \quad (\text{k1})$$

$$\Box(A \rightarrow B) \rightarrow (\Diamond A \rightarrow \Diamond B) \quad (\text{k2})$$

$$\Diamond(A \vee B) \rightarrow (\Diamond A \vee \Diamond B) \quad (\text{k3})$$

$$(\Diamond A \rightarrow \Box B) \rightarrow \Box(A \rightarrow B) \quad (\text{k4})$$

$$\neg \Diamond \perp \quad (\text{k5})$$

Two different proposals for the semantics of IML are popular [1]. The first one is called intuitionistic IK, which requires all of the above axioms. The other one is constructive IK and only includes axioms k1 and k2. This paper is built on top of the IML formulation found in [6], and thus considers intuitionistic IK.

By adding combinations of the intuitionistic modal axioms T,D,B,4, and 5 to IK 15 different logics are generated, as shown in Figure 1. All logics were proved complete and sound in [2]. The aim

of this paper is to analyse the relations between these logics as they are depicted in the cube. We were able to show that the axioms T,D,B,4, and 5 are equal to some frame correspondences. Using these correspondences we will verify that some combinations of the intuitionistic modal axioms are indeed equivalent. Furthermore, proofs are given to demonstrate that some logics are stronger than others. In summary, we will then verify the whole intuitionistic modal logic cube, except for some trivial statements.

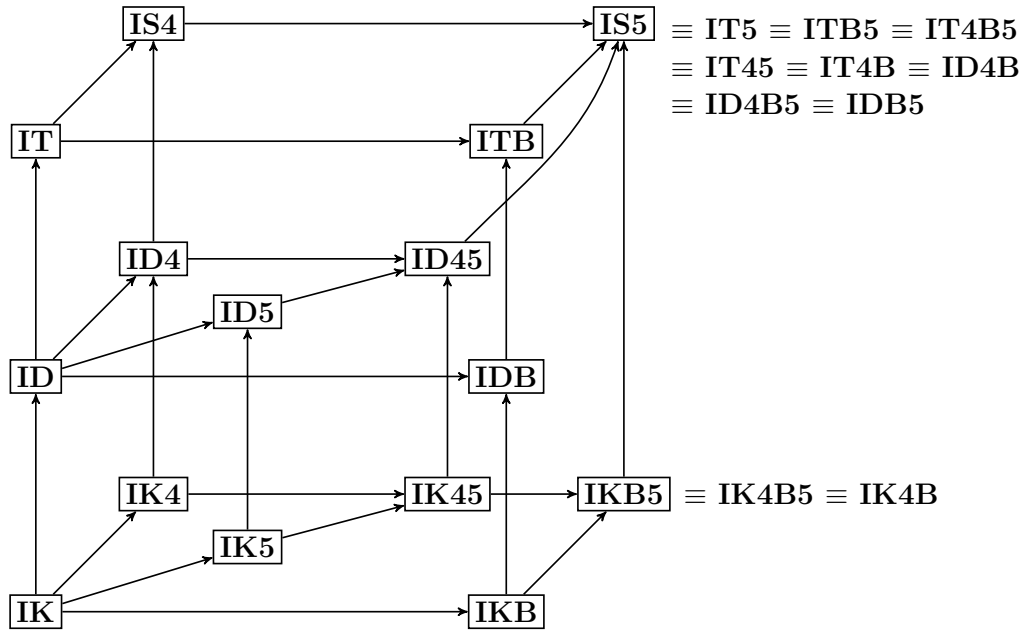


Fig. 1: IML Cube

To this end we use an embedding of IML in higher-order logic. Embeddings of various other logics in higher-order logic have already been realised before (see [3,5] and the references herein).

This paper is automatically generated from the Isabelle/HOL and is thus formally verified. Within Isabelle/HOL, we especially make extensive use of the automated reasoning tools provided by Isabelle. Sledgehammer [8] is employed to access higher-order reasoning tools LEO-II [9], Satallax [10] and Nitpick [12] for proof search.

The paper is structured as follows: Firstly, Section 2 defines syntax and semantics of IML, Section 3 presents the IML cube and gives information about frame correspondences in IML. Section 4 defines an encoding of IML in higher-order logic and Section 5 gives evidence for the correspondences named in Section 3. Section 6 shows that some different axiomatisations are equivalent whereas in Section 7 cases are shown in which one logic is stronger than another.

2 Intuitionistic Modal Logic

The language of IML is identical to the language of classical modal logic. A formula A is generated by:

$$A ::= A \wedge A \mid A \vee A \mid A \rightarrow A \mid \Diamond A \mid \Box A \mid \perp \mid q$$

where $q \in Q$ and $Q = a, b, c, \dots$ is a set of atomic sentences. $\neg A$ is defined as $A \rightarrow \perp$.

A Kripkean intuitionistic modal frame is a tuple $\langle W, \leq, R \rangle$, where W is a set of worlds and R is the modal accessibility relation. $\leq$ is called the intuitionistic information relation and is partially ordered. For convenience, the $\geq$ -notation will be used: $w \leq v$ iff $v \geq w$. An intuitionistic modal model for the language L is a tuple $\langle W, \leq, R, V \rangle$. V is a valuation function $W \rightarrow \mathcal{P}(Q)$ which maps a world to those atomic sentences which are true in this world.

Plotkin et al. [6] emphasises that the choice of frame conditions is strongly connected to the semantics of the modal operators. Four different ways are proposed how the $\leq$ -relation and the R relation may be connected. As suggested, we will use frame conditions 1 and 3.

$$\text{if } w \leq w' \text{ and } wRv \text{ then } \exists v'. w'Rv' \text{ and } v \leq v' \quad (1)$$

$$\text{if } w \leq w' \text{ and } w'Rv' \text{ then } \exists v. wRv \text{ and } v \leq v' \quad (2)$$

$$\text{if } v \leq v' \text{ and } wRv \text{ then } \exists w'. w'Rv' \text{ and } w \leq w' \quad (3)$$

$$\text{if } v \leq v' \text{ and } w'Rv' \text{ then } \exists w. wRv \text{ and } w \leq w' \quad (4)$$

To evaluate a formula A in the context of a world w an evaluation relation $\models$ is defined.

$$\begin{aligned} w \models q & \quad \text{iff } q \in V(w) \\ w \models A \wedge B & \quad \text{iff } w \models A \text{ and } w \models B \\ w \models A \vee B & \quad \text{iff } w \models A \text{ or } w \models B \\ w \models A \rightarrow B & \quad \text{iff } \forall w' \geq w \text{ if } w' \models A \text{ then } w' \models B \\ w \models \Diamond A & \quad \text{iff } \exists u. wRu \text{ and } u \models A \\ w \models \Box A & \quad \text{iff } \forall w' \geq w \forall u. \text{ if } w'Ru \text{ then } u \models A \end{aligned}$$

In addition to the frame conditions shown above the following statement applies:

$$\text{if } w \leq w' \text{ then } V(w) \subset V(w') \quad (5)$$

Lemma 1 (Monotonicity). *if $w \leq w'$ and $w \models A$ then $w' \models A$*

Proof. Using (1), (5), $\models$ and an induction on A [13].

Furthermore, a formula A is:

$$\text{valid in a model } M = \langle W, \leq, R, V \rangle \quad \text{iff } \forall w \in W : w \models A. \quad (6)$$

$$\text{valid in a frame } \langle W, \leq, R \rangle \quad \text{iff } \forall V : \langle W, \leq, R, V \rangle \models A. \quad (7)$$

$$\text{valid iff } \forall \langle W, \leq, R \rangle : \langle W, \leq, R \rangle \models A. \quad (8)$$

3 Intuitionistic Modal Logic Cube and Frame Correspondences

By adding the intuitionistic modal axioms D, T, B, 4, and 5 to IK, 15 different logics are generated. These logics can be arranged in form of a cube, as depicted in Figure 1. The modal axioms are not the same as in classical modal logic, which is due to the breakdown of duality between $\Box$ and $\Diamond$: $\Box A \leftrightarrow \neg \Diamond \neg A$ does not hold in IML. As a result of this breakdown, modal axioms consist of two parts, which are here distinguished by an annotation of either a $\Diamond$ or a $\Box$ to their name.

At this point, we could use the intuitionistic modal axioms for the verification of the IML cube. However, it is more efficient to use corresponding frame conditions, as done by Benz Müller et al. in [4] for classical modal logic. In [6] Plotkin et al. give two reasons why it would be unlikely that the intuitionistic modal axioms correspond exactly to the same frame correspondence than in classical logic.

Firstly, because of the aforementioned breakdown, it would be unlikely if both parts of an intuitionistic modal axiom correspond to the same frame condition. Secondly, a correspondence theorem should not only make restraints on R, but also on the relationship between R and $\leq$. In their argumentation Plotkin et al. do not include thoughts about the influence of the properties the $\leq$ -relation has, nor about the interconnection defined through frame condition 1 and 3. Most other authors refer to these two reasons.

When we tried to check this inequality we were surprised to see that instead of finding a counterexample we could prove that all intuitionistic modal logic axioms are indeed equivalent to their classical frame correspondence equivalent. Even after examining all our assumptions most thoroughly, we could not find any divergence to the axiomatisation of Plotkin et al. To fully understand the reasons for this behaviour further work is needed. Possibly we need to reexamine the long-established position that the classical correspondences are not valid in the setting of Plotkin et al.

Nevertheless we will work with slightly different, already established frame conditions, which make statements about the connection of the $\leq$ and R relations. Each correspondence axiom is an instance of the intuitionistic version of the $G^{k,l,m,n}$ schema [6], also called Lemmon-Scott schema.

Definition 1. $G^{k,l,m,n}$ is the schema: $\Diamond^k \Box^l A \rightarrow \Box^m \Diamond^n A$, for $k, l, m, n \geq 0$

Theorem 1. A modal frame $\langle W, \leq, R \rangle$ validates $G^{k,l,m,n}$ iff the following statement applies: if $w R^k u$ and $w R^m v$ then $\exists u' \geq u \exists x. (u' R^l x \wedge v R^n x)$.

Where R^n for $n \geq 0$ is:

$$\begin{aligned} w R^0 v &\text{ iff } w = v \\ w R^{n+1} v &\text{ iff } \exists u. w R u \text{ and } u R^n v \end{aligned}$$

By applying the schema to all IML axioms we achieved the results indicated in Table 1. The equality to the correspondences has been proven in Section 5. The classical correspondences are illustrated in the same table and the equality is also shown in Section 5.

4 An Embedding of Intuitionistic Modal Logics in HOL

An embedding of IML in higher-order logic is used to verify the IML cube. This has already been done by Benz Müller et al. [4] in the scope of the modal logic cube. In the following we assume

Table 1: Intuitionistic modal axioms D,T,B,4, and 5 with corresponding frame conditions

name	axiom	intuitionistic frame correspondence	classical correspondence
T□:	$\Box A \rightarrow A$	$\forall w. \exists u'. w \leq u' \wedge u' R w$	$\forall w. w R w$ (reflexivity)
T◇:	$A \rightarrow \Diamond A$	$\forall w. \exists u'. w \leq u' \wedge w R u'$	
B□:	$A \rightarrow \Box \Diamond A$	$\forall w u. w R u \rightarrow \exists u'. w \leq u' \wedge u R u'$	$\forall w u. w R u \rightarrow u R w$ (symmetry)
B◇:	$\Diamond \Box A \rightarrow A$	$\forall w u. w R u \rightarrow \exists u'. u \leq u' \wedge u' R w$	
D:	$\Box A \rightarrow \Diamond A$	$\forall w. \exists u'. w \leq u' \wedge (\exists x. u' R x \wedge w R x)$	$\forall w. \exists u. w R u$ (seriality)
4◇:	$\Diamond \Diamond A \rightarrow \Diamond A$	$\forall w v. (\exists u. w R u \wedge u R v) \rightarrow (\exists u'. v \leq u' \wedge w R u')$	$\forall w u v. w R u \wedge u R v \rightarrow w R v$ (transitivity)
4□:	$\Box A \rightarrow \Box \Box A$	$\forall w v. (\exists u. w R u \wedge u R v) \rightarrow (\exists u'. w \leq u' \wedge u' R v)$	
5◇:	$\Diamond A \rightarrow \Box \Diamond A$	$\forall w v u. (w R u \wedge w R v) \rightarrow (\exists u'. u \leq u' \wedge u' R v)$	$\forall w u v. w R u \wedge w R v \rightarrow u R v$ (euclideaness)
5□:	$\Diamond \Box A \rightarrow \Box A$	$\forall w v u. (w R u \wedge w R v) \rightarrow (\exists u'. u \leq u' \wedge v R u')$	

constant-domains. Modification for varying domains exist but are not further explored here see [5]. In IML, a formula is dependent on its context, called its possible world. That is why we introduce a type of possible worlds ι . Using that type we can translate an IML formula into a HOL term of type $(\iota \Rightarrow \text{bool})$, here abbreviated as σ . Additionally to the types, the $\leq$ and R relation are modelled as constants.

typed decl ι

type-synonym $\sigma = (\iota \Rightarrow \text{bool})$

consts $R :: \iota \Rightarrow \iota \Rightarrow \text{bool}$ — accessibility relation

consts $le :: \iota \Rightarrow \iota \Rightarrow \text{bool}$ — ordering relation

A new evaluation function is defined, which uses the HOL operators. To distinguish between these operators and the new lifted ones we use bold lettering.

abbreviation $\text{mand} :: \sigma \Rightarrow \sigma \Rightarrow \sigma$ **where** $A \wedge B \equiv \lambda w. A w \wedge B w$

abbreviation $\text{mor} :: \sigma \Rightarrow \sigma \Rightarrow \sigma$ **where** $A \vee B \equiv \lambda w. A w \vee B w$

abbreviation $\text{mimp} :: \sigma \Rightarrow \sigma \Rightarrow \sigma$ **where** $A \rightarrow B \equiv \lambda w. \forall w'. w le w' \longrightarrow (A w' \longrightarrow B w')$

abbreviation $\text{mbox} :: \sigma \Rightarrow \sigma$ **where** $\Box A \equiv \lambda w. \forall w' v'. w le w' \longrightarrow (w' R v' \longrightarrow A v')$

abbreviation $\text{mdia} :: \sigma \Rightarrow \sigma$ **where** $\Diamond A \equiv \lambda w. \exists v. (w R v \wedge A v)$

abbreviation $\text{mfalse} :: \sigma$ **where** $\perp \equiv \lambda w. \text{False}$

abbreviation $\text{mnot} :: \sigma \Rightarrow \sigma$ **where** $\neg A \equiv \lambda w. (A \rightarrow \perp) w$

abbreviation $\text{valid} :: \sigma \Rightarrow \text{bool}$ **where** $[p] \equiv \forall w. p w$

The $\leq$ -relation is partially ordered. We also define the conditions named in Section 2.

axiomatization where $FR: \forall w. w le w$

axiomatization where $FT: \forall w u v. (w le u \wedge u le v) \longrightarrow w le v$

axiomatization where $FA: \forall w u. (w le u \wedge u le w) \longrightarrow (w=u)$

axiomatization where $F1: \forall w w' v. (w le w' \wedge w R v) \longrightarrow (\exists v'. w' R v' \wedge v le v')$

axiomatization where $F3: \forall v v' w. (v le v' \wedge w R v) \longrightarrow (\exists w'. w' R v' \wedge w le w')$

Proposition (5) can not be translated directly into HOL because the valuation function is included in the embedding. We can use Lemma 1 (Monotonicity) instead in which (5) is included.

axiomatization where $Mon: \forall w w'. \forall A. (w le w' \wedge A w) \longrightarrow A w'$

5 Proofs of the Frame Correspondences

Using the embedding from above it is possible to validate the frame correspondences. The proofs raise awareness of the challenges automatic theorem proving still has to face. All of them were

suggested by Sledgehammer. However, nine of them could not be reconstructed. Unfortunately, one statement (lemma A4-b-2) could not be verified at all. On the one hand this is an issue for future work in Isabelle, on the other hand it should be enough for this paper to know that some proof exists.

5.1 Intuitionistic modal axioms

abbreviation *T-dia* **where** $T\text{-dia} \equiv \forall A. \lfloor (A \rightarrow \Diamond A) \rfloor$

abbreviation *T-box* **where** $T\text{-box} \equiv \forall A. \lfloor ((\Box A) \rightarrow A) \rfloor$

abbreviation *T* **where** $T \equiv \forall A. \lfloor (((\Box A) \rightarrow A) \wedge (A \rightarrow \Diamond A)) \rfloor$

abbreviation *B-dia* **where** $B\text{-dia} \equiv \forall A. \lfloor (((\Diamond \Box A) \rightarrow A)) \rfloor$

abbreviation *B-box* **where** $B\text{-box} \equiv \forall A. \lfloor (((A \rightarrow \Box \Diamond A)) \rfloor$

abbreviation *B* **where** $B \equiv \forall A. \lfloor (((\Diamond \Box A) \rightarrow A) \wedge (A \rightarrow \Box \Diamond A)) \rfloor$

abbreviation *D* **where** $D \equiv \forall A. \lfloor ((\Box A) \rightarrow \Diamond A) \rfloor$

abbreviation *IV-box* **where** $IV\text{-box} \equiv \forall A. \lfloor (((\Box A) \rightarrow \Box \Box A)) \rfloor$

abbreviation *IV-dia* **where** $IV\text{-dia} \equiv \forall A. \lfloor (((\Diamond \Diamond A) \rightarrow \Diamond A)) \rfloor$

abbreviation *IV* **where** $IV \equiv \forall A. \lfloor (((\Box A) \rightarrow \Box \Box A) \wedge ((\Diamond \Diamond A) \rightarrow \Diamond A)) \rfloor$

abbreviation *V-dia* **where** $V\text{-dia} \equiv \forall A. \lfloor (((\Diamond \Box A) \rightarrow \Box A)) \rfloor$

abbreviation *V-box* **where** $V\text{-box} \equiv \forall A. \lfloor (((\Diamond A) \rightarrow \Box \Diamond A)) \rfloor$

abbreviation *V* **where** $V \equiv \forall A. \lfloor (((\Diamond \Box A) \rightarrow \Box A) \wedge ((\Diamond A) \rightarrow \Box \Diamond A)) \rfloor$

5.2 Frame Correspondences

abbreviation *FC-T-dia* **where** $FC\text{-}T\text{-}dia \equiv \forall w. \exists u'. w \text{ le } u' \wedge w R u'$

abbreviation *FC-T-box* **where** $FC\text{-}T\text{-}box \equiv \forall w. \exists u'. w \text{ le } u' \wedge u' R w$

abbreviation *FC-T* **where** $FC\text{-}T \equiv FC\text{-}T\text{-}dia \wedge FC\text{-}T\text{-}box$

abbreviation *FC-B-dia* **where** $FC\text{-}B\text{-}dia \equiv \forall w u. w R u \longrightarrow (\exists u'. u \text{ le } u' \wedge u' R w)$

abbreviation *FC-B-box* **where** $FC\text{-}B\text{-}box \equiv \forall w u. w R u \longrightarrow (\exists u'. w \text{ le } u' \wedge u R u')$

abbreviation *FC-B* **where** $FC\text{-}B \equiv FC\text{-}B\text{-}dia \wedge FC\text{-}B\text{-}box$

abbreviation *FC-D* **where** $FC\text{-}D \equiv \forall w. \exists u'. w \text{ le } u' \wedge (\exists x. u' R x \wedge w R x)$

abbreviation *FC-IV-dia*

where $FC\text{-}IV\text{-}dia \equiv \forall w u. (\exists v. w R v \wedge v R u) \longrightarrow (\exists u'. u \text{ le } u' \wedge w R u')$

abbreviation *FC-IV-box*

where $FC\text{-}IV\text{-}box \equiv \forall w u. (\exists v. w R v \wedge v R u) \longrightarrow (\exists u'. w \text{ le } u' \wedge u' R u)$

abbreviation *FC-IV* **where** $FC\text{-}IV \equiv FC\text{-}IV\text{-}dia \wedge FC\text{-}IV\text{-}box$

abbreviation *FC-V-dia*

where $FC\text{-}V\text{-}dia \equiv \forall w v u. (w R u \wedge w R v) \longrightarrow (\exists u'. u \text{ le } u' \wedge u' R v)$

abbreviation *FC-V-box*

where $FC\text{-}V\text{-}box \equiv \forall w v u. (w R u \wedge w R v) \longrightarrow (\exists u'. u \text{ le } u' \wedge v R u')$

abbreviation *FC-V* **where** $FC\text{-}V \equiv FC\text{-}V\text{-}dia \wedge FC\text{-}V\text{-}box$

5.3 Classical Frame Correspondences

abbreviation *ref* **where** $ref \equiv \forall w. w R w$

abbreviation *ser* **where** $ser \equiv \forall w. \exists v. w R v$

abbreviation *sym* where $\text{sym} \equiv \forall w u . w R u \longrightarrow u R w$
 abbreviation *trans* where $\text{trans} \equiv \forall w u v . w R u \wedge u R v \longrightarrow w R v$
 abbreviation *eucl* where $\text{eucl} \equiv \forall w u v . w R u \wedge w R v \longrightarrow u R v$

5.4 Proof of Frame Correspondences

Axiom $T \Diamond$ corresponds to FC- $T \Diamond$:

lemma *A1-a-1*: $FC\text{-}T\text{-}dia \longrightarrow T\text{-}dia$ using *Mon* by *blast*
 lemma *A1-a-2*: $T\text{-}dia \longrightarrow FC\text{-}T\text{-}dia$ by (*meson FR*)
 theorem *A1-a*: $T\text{-}dia \longleftrightarrow FC\text{-}T\text{-}dia$ using *A1-a-1 A1-a-2* by *auto*

Axiom $T \Box$ corresponds to FC- $T \Box$:

lemma *A1-b-1*: $T\text{-}box \longrightarrow FC\text{-}T\text{-}box$ — by (*metis FR Mon ext*) sorry
 lemma *A1-b-2*: $FC\text{-}T\text{-}box \longrightarrow T\text{-}box$ by *fastforce*
 theorem *A1-b*: $FC\text{-}T\text{-}box \longleftrightarrow T\text{-}box$ using *A1-b-1* by *blast*

T corresponds to reflexivity:

lemma *A1-c-1*: $ref \longrightarrow T$ using *FR* by *blast*
 lemma *A1-c-2*: $T \longrightarrow ref$ — by (*meson FR*) sorry
 theorem *A1-c*: $T \longleftrightarrow ref$ by (*smt A1-c-1 A1-c-2*)

Axiom $B \Diamond$ corresponds to FC- $B \Diamond$:

lemma *A2-a-1*: $FC\text{-}B\text{-}dia \longrightarrow B\text{-}dia$ by *blast*
 lemma *A2-a-2*: $B\text{-}dia \longrightarrow FC\text{-}B\text{-}dia$ — by (*metis FR Mon ext*) sorry
 theorem *A2-a*: $B\text{-}dia \longleftrightarrow FC\text{-}B\text{-}dia$ using *A2-a-2* by *blast*

Axiom $B \Box$ corresponds to FC- $B \Box$:

theorem *A2-b*: $B\text{-}box \longleftrightarrow FC\text{-}B\text{-}box$ by (*meson FR Mon*)

Axiom B corresponds to symmetry:

theorem *A2-c-1*: $B \longrightarrow sym$ — by (*meson FR*) sorry
 lemma *A2-c-2*: $sym \longrightarrow B$ using *FR Mon* by *blast*
 theorem *A2-c*: $sym \longleftrightarrow B$ by (*smt A2-c-1 A2-c-2*)

Axiom D corresponds to FC-D:

theorem *A3-a*: $D \longleftrightarrow FC\text{-}D$ using *FR* by *blast*

Axiom D corresponds to seriality:

theorem *A3-b*: $D \longleftrightarrow ser$ using *FR* by *fastforce*

Axiom $IV \Diamond$ corresponds to FC- $IV \Diamond$:

theorem *A4-a*: $IV\text{-}dia \longleftrightarrow FC\text{-}IV\text{-}dia$ by (*meson FR Mon*)

Axiom $IV \Box$ corresponds to FC- $IV \Box$:

lemma *A4-b-1*: $FC\text{-}IV\text{-}box \longrightarrow IV\text{-}box$ by (*smt Mon*)
 lemma *A4-b-2*: $IV\text{-}box \longrightarrow FC\text{-}IV\text{-}box$ sorry
 theorem *A4-b*: $FC\text{-}IV\text{-}box \longleftrightarrow IV\text{-}box$ by (*smt A4-b-1 A4-b-2*)

Axiom IV corresponds to transitivity:

lemma *A4-c-1*: $trans \longrightarrow IV$ by (*meson Mon*)
 lemma *A4-c-2*: $IV \longrightarrow trans$ — by (*metis FR*) sorry

theorem *A4-c*: $IV \longleftrightarrow trans$ **using** *A4-c-1* *A4-c-2* **by** *satx*

Axiom $V\Diamond$ corresponds to $FC-V\Diamond$:

lemma *A5-a-1*: $FC-V-dia \longrightarrow V-dia$ **by** (*meson Mon*)

lemma *A5-a-2*: $V-dia \longrightarrow FC-V-dia$ **sorry**

lemma *A5*: $V-dia \longleftrightarrow FC-V-dia$ **using** *A5-a-1* *A5-a-2* **by** *blast*

Axiom $V\Box$ corresponds to $FC-V\Box$:

lemma *A5-b-1*: $FC-V-box \longrightarrow V-box$ **by** (*meson Mon*)

lemma *A5-b-2*: $V-box \longrightarrow FC-V-box$ **by** (*meson FR*)

theorem *A5-b*: $FC-V-box \longleftrightarrow V-box$ **using** *A5-b-1* *A5-b-2* **by** *blast*

Axiom V corresponds to euclideaness:

lemma *A6-c-1*: $V-box \longrightarrow eucl$ — **by** (*metis FR*) **sorry**

lemma *A6-c-2*: $eucl \longrightarrow V-box$ **by** (*meson Mon*)

lemma *A6-c-3*: $V-dia \longrightarrow eucl$ — **by** (*metis FR Mon*) **sorry**

lemma *A6-c-4*: $eucl \longrightarrow V-dia$ **by** (*meson FR Mon*)

theorem *A6-c-5*: $eucl \longleftrightarrow V$ **by** (*smt A6-c-1 A6-c-2 A6-c-4*)

6 Alternative Axiomatisations

As shown in Figure 1, in some cases the same logic can be obtained by adding different combinations of axioms. For example, the axioms used to obtain **IS5**, namely V and T , are equivalent to the ones to obtain **D4B**, which are D , IV , and B . In this Section, proofs are given to show the equality of these logics. Therefore, the correspondence axioms from Section 3 are used. This means that instead of showing $V \wedge T \equiv D \wedge V \wedge B$, it is sufficient to show $FC-V \wedge FC-T \equiv FC-D \wedge FC-IV \wedge FC-B$. To avoid redundancy, the helper lemmata defined in the next Subsection are defined. For H3 and H4 Sledgehammer found a proof but metis could not reconstruct them. This means that each proof which is based on H3 or H4 can not be reconstructed either.

Helper lemmata

lemma *H1*: $FC-V \wedge FC-T \longrightarrow FC-B$ **by** (*meson FR Mon*)

lemma *H2*: $FC-V \wedge FC-T \longrightarrow FC-D$ **using** *FR* **by** *blast*

lemma *H3*: $FC-B \wedge FC-V \longrightarrow FC-IV$ — **by** (*metis Mon*) **sorry**

lemma *H4*: $FC-B \wedge FC-IV \longrightarrow FC-V$ — **by** (*metis FR Mon*) **sorry**

lemma *H5*: $FC-B \wedge FC-IV \wedge FC-D \longrightarrow FC-T$ **using** *H4* **by** *meson*

IT5 $\iff$ ITB5:

theorem *B1*: $(FC-B \wedge FC-T \wedge FC-V) \longleftrightarrow (FC-T \wedge FC-V)$ **using** *H1* **by** *fastforce*

IT5 $\iff$ IT45:

theorem *B2*: $(FC-T \wedge FC-V \wedge FC-IV) \longleftrightarrow (FC-T \wedge FC-V)$ **using** *H3 H1* **by** *fastforce*

IT5 $\iff$ IT4B5:

theorem *B3*: $(FC-T \wedge FC-V) \longleftrightarrow (FC-T \wedge FC-IV \wedge FC-B \wedge FC-V)$ **using** *B1 B2* **by** *satx*

IT5 $\iff$ IT4B:

theorem *B4*: $(FC-T \wedge FC-IV \wedge FC-B) \longleftrightarrow (FC-T \wedge FC-V)$ **using** *H4 B1 B2* **by** *satx*

IT5 $\iff$ ID4B:

theorem B5: $(FC-B \wedge FC-IV \wedge FC-D) \longleftrightarrow (FC-T \wedge FC-V)$ **using** $H1\ H2\ H3\ H4\ H5$ **by** *satx*

IT5 $\iff$ ID4B5:

theorem B6: $(FC-T \wedge FC-V) \longleftrightarrow (FC-B \wedge FC-IV \wedge FC-D \wedge FC-V)$ **using** $B5\ H2$ **by** *satx*

IT5 $\iff$ IDB5:

theorem B7: $(FC-B \wedge FC-D \wedge FC-V) \longleftrightarrow (FC-T \wedge FC-V)$ **using** $H3\ H5\ B6$ **by** *satx*

IKB5 $\iff$ IK4B5:

theorem B8: $(FC-IV \wedge FC-B \wedge FC-V) \longleftrightarrow (FC-B \wedge FC-V)$ **using** $H3$ **by** *satx*

IKB5 $\iff$ IK4B:

theorem B9-a: $(FC-IV \wedge FC-B) \longleftrightarrow (FC-B \wedge FC-V)$ **using** $H3\ H4$ **by** *satx*

7 Inclusion Relations

In the previous Sections we proved that some logics are actually the same. Now we want to show which logics differ. Analogous to [4] this paper concentrates on the backward direction of an edge within the IML cube. The forward direction is always trivial. For example, to show that each theorem of ID45 is a theorem of logic ID5 it is enough to omit axiom 4. Thus, we want to examine whether there are theorems of ID5 that can not be proved in ID45. The notation $A > B$ is used to indicate that in logic A strictly more theorems are provable than in logic B.

The following methodology is mostly adopted from [4]. That paper names several steps which are applied to all edges in the cube. Only one step was omitted because it was rarely possible to prove it.

7.1 Preparation

The following three abbreviations are taken over from [4] directly. They are needed because some logics are only equivalent if the model considered has enough worlds. For example, *two-world-model* forces that there are at least two worlds and that they are not equal. The idea behind Benzmlüllers et al. methodology is to determine the minimum number of worlds which fulfil an inclusion relation.

abbreviation *one-world-model* :: $\iota \Rightarrow \text{bool}$ **where** $\#^1\ w1 \equiv \forall x. x = w1$

abbreviation *two-world-model* :: $\iota \Rightarrow \iota \Rightarrow \text{bool}$ **where** $\#^2\ w1\ w2 \equiv (\forall x. x = w1 \vee x = w2) \wedge w1 \neq w2$

abbreviation *three-world-model* :: $\iota \Rightarrow \iota \Rightarrow \iota \Rightarrow \text{bool}$ **where** $\#^3\ w1\ w2\ w3 \equiv (\forall x. x = w1 \vee x = w2 \vee x = w3) \wedge w1 \neq w2 \wedge w1 \neq w3 \wedge w2 \neq w3$

Additionally, we define three constants of type world. They are used later as arguments for the world-model operators above.

consts $i1::\iota\ i2::\iota\ i3::\iota$

To improve the readability the following abbreviations for the logics are defined:

abbreviation *IT5* **where** $IT5 \equiv FC-V \wedge FC-T$

abbreviation *IKB5* **where** $IKB5 \equiv FC-B \wedge FC-V$

abbreviation *IK4* **where** $IK4 \equiv FC-IV$

abbreviation *IKB* **where** $IKB \equiv FC-B$

abbreviation *IK5* **where** $IK5 \equiv FC-V$

abbreviation *IK45* **where** $IK45 \equiv FC-IV \wedge FC-V$

abbreviation ID **where** $ID \equiv FC-D$
abbreviation IDB **where** $IDB \equiv FC-D \wedge FC-B$
abbreviation ID_4 **where** $ID_4 \equiv FC-D \wedge FC-IV$
abbreviation IT **where** $IT \equiv FC-T$
abbreviation IT_4 **where** $IT_4 \equiv FC-T \wedge FC-IV$
abbreviation ID_5 **where** $ID_5 \equiv FC-D \wedge FC-V$
abbreviation ID_{45} **where** $ID_{45} \equiv FC-D \wedge FC-IV \wedge FC-V$
abbreviation ITB **where** $ITB \equiv FC-T \wedge FC-B$

7.2 Step A

We need to obtain the information how many worlds a model has to have to make the inclusion relation true. Therefore, we assume that a false inclusion is valid and apply Nitpick on it. Nitpick will first try to test models with one world, then with two, and so on.

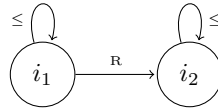
In the $IKB > IK$ example, nitpick can not find a counterexample with one world: For all models, containing only one world, the statement IKB is true. To retrieve the arity information when countermodels are possible, we assumed that each theorem in IKB is also a theorem in IK :

lemma C1-a: IKB

All lemmata produced by applying step A are named $C*-a$. Nitpick generates a countermodel for each of them, 16 are countermodels with two worlds, 7 with one world and 1 with three worlds. For the $IKB > IK$ example the countermodel found by Nitpick is:

$$\begin{aligned}
 R &= (\lambda x. -) ((\iota_1, \iota_1) := False, (\iota_1, \iota_2) := True, (\iota_2, \iota_1) := False, (\iota_2, \iota_2) := False) \\
 le &= (\lambda x. -) ((\iota_1, \iota_1) := True, (\iota_1, \iota_2) := False, (\iota_2, \iota_1) := False, (\iota_2, \iota_2) := True)
 \end{aligned}$$

This can be represented as a diagram:



$IKB > IK$ may be valid only for models with two or more worlds because nitpick found a counterexample for the statement $IKB \not\geq IK$ with two worlds.

7.3 Step B

Knowing how many worlds are needed we can use the abbreviations defined in Section 7.1 to enforce this number of worlds. In [4] it was possible to use the arity information directly. But when we tried to prove:

$$\#^2 i1 i2 \rightarrow \neg IKB$$

Nitpick found a counterexample. It seems that inclusion is dependent on properties of R and the $\leq$ -relation. Anyway, in [4] the Metis-based integration into Isabelle also failed in a few cases. For these cases the authors took another method and used all information Nitpick gave them in its

countermodel. In this paper we will always utilise all the information about R and the $\leq$ -relation. In the $IKB > IK$ example, we know for example that $i1 \leq i1$ is true, whereas $i1 \leq i2$ is false.

lemma C1-b: $\#^2 i1 i2 \wedge \neg i1 R i1 \wedge i1 R i2 \wedge \neg i2 R i1 \wedge \neg i2 R i2 \wedge i1 le i1$
 $\wedge \neg i1 le i2 \wedge \neg i2 le i1 \wedge i2 le i2 \rightarrow \neg IKB$

We named the resulting theorems uniformly C*-C.

In nearly all cases the Metis-based integration into Isabelle failed. However, every theorem except one (lemma C5-e) could be proven by CVC4 instead. With this step we show that the links in the intuitionistic modal cube are indeed correct.

7.4 Step C

Although the inclusions are already shown at this point, we want to determine whether the countermodels Nitpick produced have the minimal number of worlds. In the $IKB > IK$ example we know that a minimum of two worlds is needed. Now we would prove:

lemma C1-c: $\#^1 i1 \rightarrow IKB$

to show that having only one world is not enough. Indeed, in that case the statement $IKB > IK$ is false for all relations R and $\leq$.

The resulting theorems are uniformly named C*-c. If a counterexample consists of one possible world only, it is not necessary to apply this step.

7.5 Proofs of Inclusion Relations

Only a few inclusion relations are shown here as examples. The remaining proofs can be found in the annex of this paper.

ID4 > IK4:

lemma C9-a: $IK4 \rightarrow ID4$ **nitpick oops**

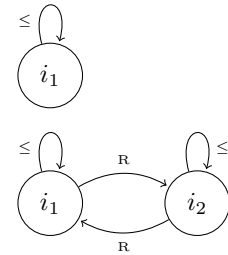
theorem C9-b: $(\#^1 i1 \wedge \neg i1 R i1 \wedge i1 le i1) \rightarrow \neg (IK4 \rightarrow ID4)$ **by smt**

IK4 > IK:

lemma C1-a: $IK4$ **nitpick oops**

theorem C1-b: $(\#^2 i1 i2 \wedge i1 le i1 \wedge i2 le i2 \wedge \neg i1 le i2 \wedge \neg i2 le i1$
 $\wedge \neg i1 R i1 \wedge i1 R i2 \wedge i2 R i1 \wedge \neg i2 R i2) \rightarrow \neg IK4$ **by smt**

lemma C1-c: $\#^1 i1 \rightarrow IK4$ **by (smt FR)**

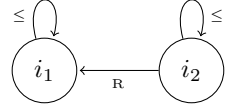


IK5 > IK:

lemma C2-a: *IK5 nitpick oops*

theorem C2-b: $(\#^2 i1 i2 \wedge \neg i1 R i1 \wedge \neg i1 R i2 \wedge i2 R i1 \wedge \neg i2 R i2 \wedge i1 le i1 \wedge \neg i1 le i2 \wedge \neg i2 le i1 \wedge i2 le i2) \rightarrow \neg IK5$ **by** *smt*

lemma C2-c: $\#^1 i1 \rightarrow IK5$ **by** *(smt FR)*

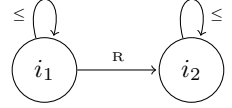


IKB > IK:

lemma C3-a: *IKB nitpick oops*

theorem C3-b: $(\#^2 i1 i2 \wedge \neg i1 R i1 \wedge i1 R i2 \wedge \neg i2 R i1 \wedge \neg i2 R i2 \wedge i1 le i1 \wedge \neg i1 le i2 \wedge \neg i2 le i1 \wedge i2 le i2) \rightarrow \neg IKB$ **by** *smt*

lemma C3-c: $\#^1 i1 \rightarrow IKB$ **by** *(smt FR)*

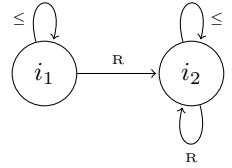


IKB5 > IK45:

lemma C7-a: *IK45 → IKB5 nitpick oops*

theorem C7-b: $(\#^2 i1 i2 \wedge \neg i1 R i1 \wedge i1 R i2 \wedge \neg i2 R i1 \wedge i2 R i2 \wedge i1 le i1 \wedge \neg i1 le i2 \wedge \neg i2 le i1 \wedge i2 le i2) \rightarrow \neg (IK45 \rightarrow IKB5)$ **by** *smt*

lemma C7-c: $\#^1 i1 \rightarrow (IK45 \rightarrow IKB5)$ **by** *(smt FR)*

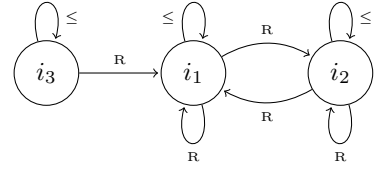


IK45 > IK5:

lemma C5-a: *IK5 → IK45 nitpick oops*

lemma C5-d: $(\#^3 i1 i2 i3 \wedge i1 R i1 \wedge i2 R i2 \wedge \neg i3 R i3 \wedge i1 R i2 \wedge i2 R i1 \wedge \neg i2 R i3 \wedge \neg i3 R i2 \wedge \neg i1 R i3 \wedge i3 R i1 \wedge i1 le i1 \wedge i2 le i2 \wedge i3 le i3 \wedge \neg i1 le i2 \wedge \neg i2 le i1 \wedge \neg i2 le i3 \wedge \neg i3 le i2 \wedge \neg i1 le i3 \wedge \neg i3 le i1) \rightarrow \neg (IK5 \rightarrow IK45)$ **by** *smt*

lemma C5-e: $(\#^2 i1 i2) \rightarrow (IK5 \rightarrow IK45)$ **sorry**



8 Conclusion

In this paper the intuitionistic modal logic cube was successfully verified. For this purpose, an embedding of IML in HOL was presented and used to show alternative axiomatisations and inclusion relationships. In the verification process, we used the counterexample generator Nitpick in combination with calls to the higher-order theorem provers LEO-II and Satallax and the SMT solver CVC4 via Isabelle's Sledgehammer tool. Our work motivates further work on the integration of LEO-II and Satallax in Isabelle.

One of the most surprising findings of this paper is that it was possible to show an equivalence between the classical frame correspondences and the intuitionistic modal axioms! It remains an open issue to find the reasons for this behaviour.

In summary, it is possible to verify all relationships in the modal logic cube. The methodology proposed in [4] could also be used to verify other cubes.

References

1. Arisaka, R., Das A., Strassburger, L., On nested Sequents for Constructive Modal Logics. In: Logical Methods in Computer Science. pp.1-33. doi: 10.1007/978-3-642-37075-5_14
2. Simpson, A.K., The proof theory and semantics of intuitionistic modal logic. University of Edinburgh, College of Science and Engineering, School of Informatics. (1994)
3. Benzmüller, C., Verifying the modal logic cube is an easy task (for higher-order automated reasoners). In: Verification, Induction, Termination Analysis. pp. 117-128. Springer, Berlin Heidelberg (2010). doi: 10.1007/978-3-642-17172-7_7
4. Benzmüller, C., Claus, M., Sultana, N., Systematic verification of the modal logic cube in Isabelle/HOL. In: Electronic Proceedings in Theoretical Computer Science, pp.27-41. doi: 10.4204/EPTCS.186.5
5. Benzmüller, C., Paleo, B. W., Higher-Order Modal Logics: Automation and Applications. In: Faber W., Paschke A. (eds) Reasoning Web. Web Logic Rules. Reasoning Web 2015. Lecture Notes in Computer Science, vol 9203 pp. 32-74. Springer, Cham (2015). doi: 10.1007/978-3-319-21768-0_2
6. Plotkin, G., Stirling, C., A framework for intuitionistic modal logic. In: Proceedings of the 1986 Conference on Theoretical aspects of reasoning about knowledge, pp. 399-406. Morgan Kaufmann Publishers Inc., Burlington, Massachusetts (1986). doi: 10.2307/2274560
7. Nipkow, T., Paulson, L.C., Wenzel, M., Isabelle/HOL: a proof assistant for higher-order logic. Springer, Berlin Heidelberg (2002). doi: 10.1007/3-540-45949-9
8. Paulson, L.C., Blanchette, J.C., Three years of experience with Sledgehammer, a practical link between automatic and interactive theorem provers. In: Sutcliffe, G., Schulz, S., Ternovska, E. (eds) IWIL 2010. The 8th International Workshop on the Implementation of Logics, vol 2. pp. 1-11, (2010)
9. Benzmüller, C., Paulson, L.C., Theiss, F., Fietzke, A., LEO-II - A cooperative automatic theorem prover for classical higher-order logic (system description). In: International Joint Conference on Automated Reasoning, pp. 162-170. Springer, Berlin Heidelberg (2008). doi: 10.1007/978-3-540-71070-7_14
10. Brown, C.E., Satallax: An automatic higher-order prover. In: International Joint Conference on Automated Reasoning, pp.111-117. Springer, Berlin Heidelberg (2012). doi: 10.1007/978-3-642-31365-3_11
11. Hurd, J., First-order proof tactics in higher-order logic theorem provers. In: Design and Application of Strategies/Tactics in Higher Order Logics, pp. 56-68. NASA Technical Reports NASA/CP-2003-212448 (2003).
12. Blanchette, J.C., Nipkow, T., Nitpick: A counterexample generator for higher-order logic based on a relational model finder. In: International Conference on Interactive Theorem Proving, pp. 131-146. Springer, Berlin Heidelberg (2010). doi: 10.1007/978-3-642-14052-5_11
13. Straßburger, L., Cut elimination in nested sequents for intuitionistic modal logics. In: International Conference on Foundations of Software Science and Computational Structures, pp. 209-224. Springer, Rome (2013). doi: 10.1007/978-3-642-37075-5_14

9 Appendix

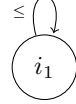
9.1 Further proofs of Inclusion Relations

The following proofs are given to complete the verification of inclusion relationships started in section 7.5. Together with the statements proven there all inclusion relationships of the intuitionistic modal logic cube are validated.

ID5 > IK5:

lemma C10-a: $IK5 \rightarrow ID5$ **nitpick oops**

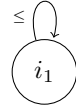
theorem C10-b: $(\#^1 i1 \wedge \neg i1 R i1 \wedge i1 le i1) \rightarrow \neg (IK5 \rightarrow ID5)$ **by smt**



ID45 > IK45:

lemma C11-a: $IK45 \rightarrow ID45$ **nitpick oops**

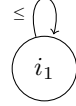
theorem C11-b: $(\#^1 i1 \wedge \neg i1 R i1 \wedge i1 le i1) \rightarrow \neg (IK45 \rightarrow ID45)$ **by smt**



IDB > IKB:

lemma C12-a: $IKB \rightarrow IDB$ **nitpick oops**

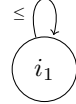
theorem C12-b: $(\#^1 i1 \wedge \neg i1 R i1 \wedge i1 le i1) \rightarrow \neg (IKB \rightarrow IDB)$ **by smt**



IS5 > IKB5:

lemma C13-a: $IKB5 \rightarrow IT5$ **nitpick oops**

theorem C13-b: $(\#^1 i1 \wedge \neg i1 R i1 \wedge i1 le i1) \rightarrow \neg (IKB5 \rightarrow IT5)$ **by smt**

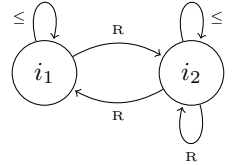


ID4 > ID:

lemma C14-a: $ID \rightarrow ID4$ **nitpick oops**

theorem C14-b: $(\#^2 i1 i2 \wedge \neg i1 R i1 \wedge i1 R i2 \wedge i2 R i1 \wedge \neg i2 R i2 \wedge i1 le i1 \wedge \neg i1 le i2 \wedge \neg i2 le i1 \wedge i2 le i2) \rightarrow \neg (ID \rightarrow ID4)$ **by smt**

lemma C14-c: $\#^1 i1 \rightarrow (ID \rightarrow ID4)$ **by (smt FR)**

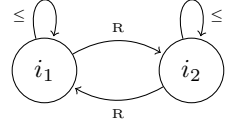


ID5 > ID:

lemma C15-a: $ID \rightarrow ID5$ **nitpick oops**

theorem C15-b: $(\#^2 i1 i2 \wedge \neg i1 R i1 \wedge i1 R i2 \wedge i2 R i1 \wedge \neg i2 R i2 \wedge i1 le i1 \wedge \neg i1 le i2 \wedge \neg i2 le i1 \wedge i2 le i2) \rightarrow \neg (ID \rightarrow ID5)$ **by smt**

lemma C15-c: $\#^1 i1 \rightarrow (ID \rightarrow ID5)$ **by (smt FR)**

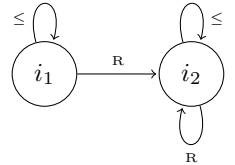


IDB > ID:

lemma C16-a: $ID \rightarrow IDB$ **nitpick oops**

theorem C16-b: $(\#^2 i1 i2 \wedge \neg i1 R i1 \wedge i1 R i2 \wedge \neg i2 R i1 \wedge i2 R i2 \wedge i1 le i1 \wedge \neg i1 le i2 \wedge \neg i2 le i1 \wedge i2 le i2) \rightarrow \neg (ID \rightarrow IDB)$ **by smt**

lemma C16-c: $\#^1 i1 \rightarrow (ID \rightarrow IDB)$ **by smt**

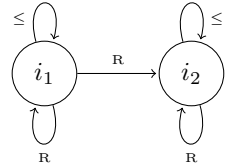


ID45 > ID4:

lemma C17-a: $ID4 \rightarrow ID45$ **nitpick oops**

theorem C17-b: $(\#^2 i1 i2 \wedge i1 R i1 \wedge i1 R i2 \wedge \neg i2 R i1 \wedge i2 R i2 \wedge i1 le i1 \wedge \neg i1 le i2 \wedge \neg i2 le i1 \wedge i2 le i2) \rightarrow \neg (ID4 \rightarrow ID45)$ **by smt**

lemma C17-c: $\#^1 i1 \rightarrow (ID4 \rightarrow ID45)$ **by (smt FR)**

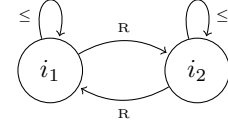


IT > I:

lemma C18-a: $ID \rightarrow IT$ **nitpick oops**

theorem C18-b: $(\#^2 i1\ i2 \wedge \neg i1\ R\ i1 \wedge i1\ R\ i2 \wedge i2\ R\ i1 \wedge \neg i2\ R\ i2 \wedge i1\ le\ i1 \wedge \neg i1\ le\ i2 \wedge \neg i2\ le\ i1 \wedge i2\ le\ i2) \rightarrow \neg (ID \rightarrow IT)$ **by smt**

lemma C18-c: $\#^1 i1 \rightarrow (ID \rightarrow IT)$ **by smt**

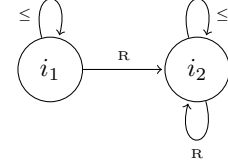


IS4 > ID4:

lemma C19-a: $ID4 \rightarrow IT4$ **nitpick oops**

theorem C19-b: $(\#^2 i1\ i2 \wedge \neg i1\ R\ i1 \wedge i1\ R\ i2 \wedge \neg i2\ R\ i1 \wedge i2\ R\ i2 \wedge i1\ le\ i1 \wedge \neg i1\ le\ i2 \wedge \neg i2\ le\ i1 \wedge i2\ le\ i2) \rightarrow \neg (ID4 \rightarrow IT4)$ **by smt**

lemma C19-c: $\#^1 i1 \rightarrow (ID4 \rightarrow IT4)$ **by smt**

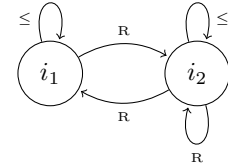


IS5 > ID45:

lemma C20-a: $ID45 \rightarrow IT5$ **nitpick oops**

theorem C20-b: $(\#^2 i1\ i2 \wedge \neg i1\ R\ i1 \wedge i1\ R\ i2 \wedge \neg i2\ R\ i1 \wedge i2\ R\ i2 \wedge i1\ le\ i1 \wedge \neg i1\ le\ i2 \wedge \neg i2\ le\ i1 \wedge i2\ le\ i2) \rightarrow \neg (ID45 \rightarrow IT5)$ **by smt**

lemma C20-c: $\#^1 i1 \rightarrow (ID45 \rightarrow IT5)$ **by smt**

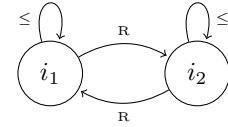


IB > IDB:

lemma C21-a: $IDB \rightarrow ITB$ **nitpick oops**

theorem C21-b: $(\#^2 i1\ i2 \wedge \neg i1\ R\ i1 \wedge i1\ R\ i2 \wedge i2\ R\ i1 \wedge \neg i2\ R\ i2 \wedge i1\ le\ i1 \wedge \neg i1\ le\ i2 \wedge \neg i2\ le\ i1 \wedge i2\ le\ i2) \rightarrow \neg (IDB \rightarrow IT5)$ **by smt**

lemma C21-c: $\#^1 i1 \rightarrow (IDB \rightarrow ITB)$ **by smt**

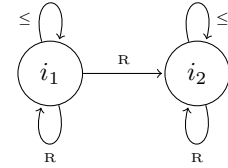


IB > IT:

lemma C22-a: $IT \rightarrow IKB$ **nitpick oops**

theorem C22-b: $(\#^2 i1\ i2 \wedge i1\ R\ i1 \wedge i1\ R\ i2 \wedge \neg i2\ R\ i1 \wedge i2\ R\ i2 \wedge i1\ le\ i1 \wedge \neg i1\ le\ i2 \wedge \neg i2\ le\ i1 \wedge i2\ le\ i2) \rightarrow \neg (IT \rightarrow IKB)$ **by smt**

lemma C22-c: $\#^1 i1 \rightarrow (IT \rightarrow IKB)$ **by smt**

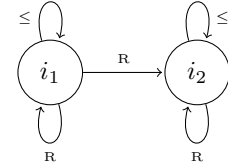


IS5 > IS4:

lemma C23-a: $IT4 \rightarrow IT5$ **nitpick oops**

theorem C23-b: $(\#^2 i1\ i2 \wedge i1\ R\ i1 \wedge i1\ R\ i2 \wedge \neg i2\ R\ i1 \wedge i2\ R\ i2 \wedge i1\ le\ i1 \wedge \neg i1\ le\ i2 \wedge \neg i2\ le\ i1 \wedge i2\ le\ i2) \rightarrow \neg (IT4 \rightarrow IT5)$ **by smt**

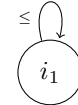
lemma C23-c: $\#^1 i1 \rightarrow (IT4 \rightarrow IT5)$ **by smt**



IS5 > IKB:

lemma C24-a: $IKB \rightarrow IT5$ **nitpick oops**

theorem C24-b: $(\#^1 i1 \wedge \neg i1\ R\ i1 \wedge i1\ le\ i1) \rightarrow \neg (IKB \rightarrow IT5)$ **by smt**

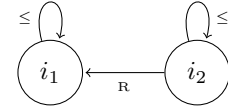


IK45 > IK4:

lemma C4-a: $IK4 \rightarrow IK45$ **nitpick oops**

theorem C4-b: $(\#^2 i1\ i2 \wedge \neg i1\ R\ i1 \wedge \neg i1\ R\ i2 \wedge i2\ R\ i1 \wedge \neg i2\ R\ i2 \wedge i1\ le\ i1 \wedge \neg i1\ le\ i2 \wedge \neg i2\ le\ i1 \wedge i2\ le\ i2) \rightarrow \neg (IK4 \rightarrow IK45)$ **by smt**

lemma C4-c: $\#^1 i1 \rightarrow (IK4 \rightarrow IK45)$ **by smt**

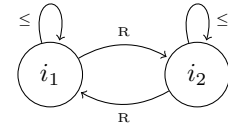


IKB5 > IKB:

lemma C6-a: $IKB \rightarrow IKB5$ **nitpick oops**

theorem C6-b: $(\#^2 i1\ i2 \wedge \neg i1\ R\ i1 \wedge i1\ R\ i2 \wedge i2\ R\ i1 \wedge \neg i2\ R\ i2 \wedge i1\ le\ i1 \wedge \neg i1\ le\ i2 \wedge \neg i2\ le\ i1 \wedge i2\ le\ i2) \rightarrow \neg (IKB \rightarrow IKB5)$ **by smt**

lemma C6-c: $\#^1 i1 \rightarrow (IKB \rightarrow IKB5)$ **by (smt FR)**



ID > IK:

lemma *C8-a: ID nitpick oops*

theorem *C8-c: $(\#^1\ i1 \wedge \neg\ i1\ R\ i1 \wedge i1\ le\ i1) \longrightarrow \neg\ ID$ by smt*

